LAB 2

Two active IP addresses:

1. 192.168.11.1
2. 192.168.11.100

Part 3: Scan Open Ports

Three open ports:

- 53 → domain
- 80 → http
- 443 → https

Part 4: Service Detection

- **Service:** DNS
- **Version:** generic response (SERVFAIL)

Part 5: OS Detection

- **Detected OS:** Linux (version 3.x – 4.x)

Part 6: Aggressive Scan

The scan revealed:

1. Presence of web services (HTTP/HTTPS)
2. SSL certificate (ZTE)
3. Linux-based system
4. Active UPnP service

Analysis Questions

1. **Number of detected machines:** 5 machines
2. **Most common ports:**
    - 80 (HTTP)

    - 443 (HTTPS)
    - 53 (DNS)

3. **Why are open ports a risk?**
   Open ports are a risk because they allow attackers to access services and can be exploited to launch attacks.

4. **What information can an attacker obtain?**

- active services
- open ports
- operating system
- potential vulnerabilities

Security Observations

- Open ports represent entry points
- Web services can be targeted
- Linux systems may contain vulnerabilities if not updated

Conclusion

This lab allowed me to understand:

- network reconnaissance
- the use of Nmap
- the risks associated with open ports